S.No	Course code	Year - semester	Subject	T	P	C
1	R23ECE-MN2241	2-2	Foundations of Quantum Computing: Physics, Engineering, and Mathematics Computing	3	0	3

Course Objectives:

- To develop a strong foundation in linear algebra, essential for quantum states and operators.
- To understand probability and statistics for quantum measurement and noise analysis.
- To explore Hamiltonian and Lagrangian mechanics as a bridge to quantum mechanics.
- To introduce electromagnetic theory relevant to quantum waveguides and optical systems.
- To study classical computing and digital logic as a stepping stone to quantum circuits.

Course Outcomes:

After successful completion of the course, the student should be able to:

- Apply concepts of vector spaces, matrices, and eigenvalue problems to quantum system modeling. (L3)
- Analyze statistical behavior of qubits using probability and interpret quantum measurement outcomes. (L3)
- Apply classical mechanics principles (Hamiltonian/Lagrangian) to understand quantum evolution. (L3)
- Explain EM wave propagation and field quantization for quantum hardware applications. (L2)
- Compare classical digital logic and architecture with quantum gates and processors. (L3)

UNIT – I: Mathematical Foundations for Quantum Systems

- Vector spaces, basis, and dimension
- Inner product, orthogonality, Hilbert spaces
- Complex matrices and linear transformations
- Eigen values, eigenvectors, and diagonalization
- Applications to quantum state representation and gate operations

Learning Outcomes:

- Understand how quantum states are represented in Hilbert space using linear algebra.
- Compute eigenvalues/eigenvectors for describing quantum observables.

UNIT – II: Probability, Statistics, and Classical Mechanics Foundations

- Descriptive statistics and probability basics
- Random variables and common distributions
- Lagrangian and Hamiltonian formulations
- Relation between classical Hamiltonians and quantum operators

Learning Outcomes:

- Apply statistical models to interpret qubit state probabilities and measurement results.
- Describe the transition from classical to quantum systems using Hamiltonian formalism.

UNIT – III: Electromagnetic Theory in Quantum Context

- Maxwell's equations (time and phasor domain)
- Wave propagation in dielectric and conducting media
- EM waves in optical fibers and waveguides
- Energy transport and quantization of EM waves

Learning Outcomes:

- Analyze EM wave behavior in classical and quantum transmission media.
- Relate classical EM theory to quantum field quantization in optics and waveguides.

UNIT – IV: Computer Architecture: Classical vs Quantum

- Von Neumann architecture and CPU design
- Memory hierarchy and instruction cycles
- Comparison with quantum computer models (control and memory)
- Limitations of classical systems for quantum operations

Learning Outcomes:

- Explain the working of classical processors and memory systems.
- Differentiate between classical and quantum architecture principles.

UNIT – V: Digital Logic and Circuit Fundamentals for Quantum Systems

- Number systems and data representation
- Basic logic gates and Boolean algebra
- Combinational circuit design
- Reversible logic and its relevance to quantum computation

Learning Outcomes:

- Design basic digital logic circuits and understand their role in classical computing.
- Explain the need for reversible logic in quantum gates and circuits.

Text Books

1. **Bernard Kolman & David Hill**, *Elementary Linear Algebra with Applications*, Pearson, 2013.
2. **Ron Larson**, *Elementary Statistics: Picturing the World*, 8th Edition, Pearson, 2023.

Reference Books

1. **Herbert Goldstein**, *Classical Mechanics*, 3rd Edition, Addison Wesley.
2. **D.J. Griffiths**, *Introduction to Electrodynamics*, 4th Edition, Cambridge University Press, 2020.
3. **M. Morris Mano**, *Computer System Architecture*, Prentice Hall.
4. **Thomas L. Floyd**, *Digital Fundamentals*, 11th Edition, Pearson.
5. **M. Morris Mano**, *Digital Logic and Computer Design*, Pearson.